Evaluation Metrics for Ordinal Models

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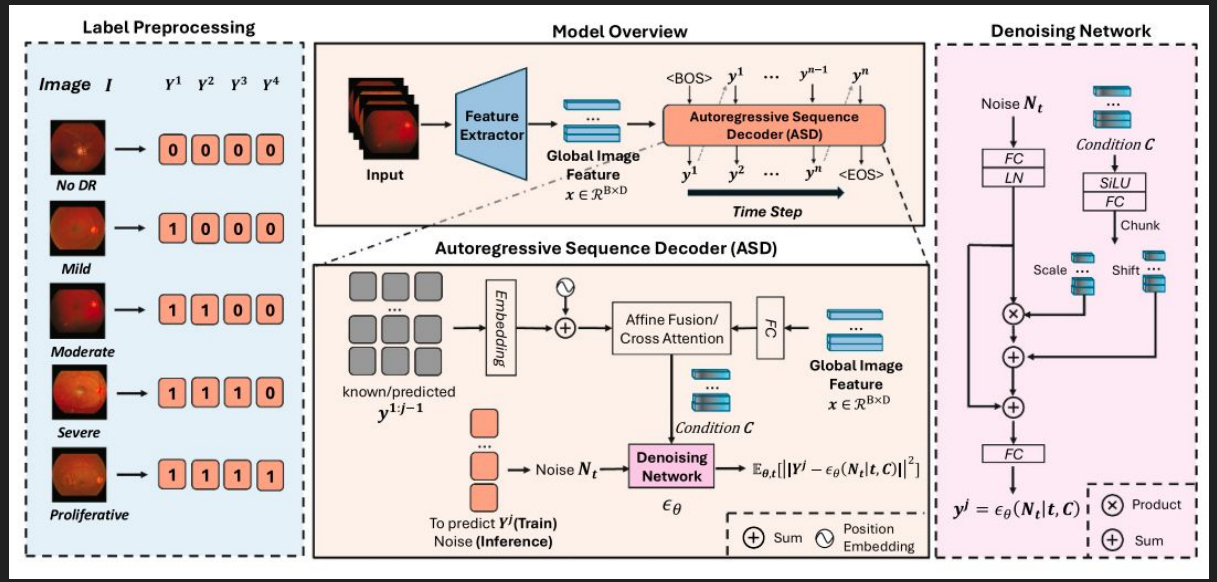
1. Motivation

1. Motivation

Parameterized Diffusion Optimization Enabled Autoregressive Ordinal Regression for Diabetic Retinopathy Grading



Medical Image Computing and Computer Assisted Intervention – MICCAI 2025



1. Motivation

Model	APTOS				Messidor				DDR				DEEPDR			
	ACC	*F1*	*Sen*	*Spec*	*ACC*	*F1*	*Sen*	*Spec*	*ACC*	*F1*	*Sen*	*Spec*	*ACC*	*F1*	*Sen*	*Spec*
ViT-B/16	69.9	38.3	36.4	68.6	63.9	41.9	43.2	72.7	57.4	26.0	22.3	63.6	62.5	39.1	39.8	71.7
CORAL* [2]	66.4	30.4	27.3	67.8	62.2	18.7	29.0	72.3	54.1	26.6	24.7	67.2	57.7	37.3	36.9	71.2
CORN* [18]	69.6	31.4	27.9	60.2	62.2	18.6	24.9	72.3	54.7	26.3	23.9	66.5	56.2	31.4	28.6	71.1
PoEs* [13]	67.2	32.4	30.4	68.2	44.7	28.1	31.8	71.2	51.3	28.0	41.6	68.8	55.5	31.3	34.2	71.8
GoL* [10]	74.2	52.2	60.7	71.4	60.8	33.8	41.4	72.6	63.6	40.9	46.1	69.3	62.7	51.2	56.7	72.8
Ord2Seq [20]	70.2	30.4	27.7	64.8	**64.7**	34.1	44.5	72.6	64.7	34.1	36.9	65.5	61.2	43.2	52.4	71.3
CLIP-DR* [23]	71.8	45.5	44.5	71.1	63.3	36.8	45.8	72.7	68.4	45.6	50.5	69.2	64.5	50.6	57.1	72.5
Ours (Affine Fusion)	74.5	52.5	58.2	71.0	58.3	37.6	**52.2**	72.7	**70.3**	**46.5**	**51.6**	68.3	**65.7**	50.5	47.6	72.5
Ours (Cross Attention)	**77.1**	**58.6**	**62.6**	**71.9**	62.5	**45.1**	50.6	**73.0**	65.0	43.9	49.5	**69.7**	65.0	**51.9**	**57.7**	**72.9**

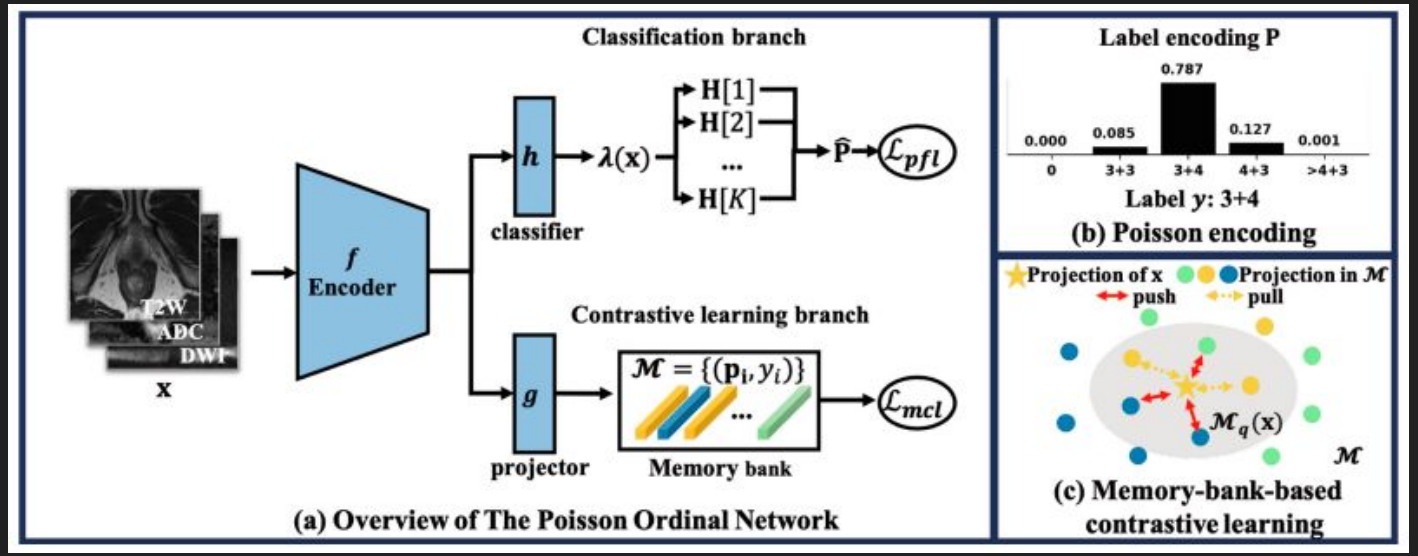
1. Motivation

Poisson Ordinal Network for Gleason Group Estimation Using Bi-Parametric MRI



Medical Image Computing and Computer Assisted Intervention – MICCAI 2024

(MICCAI 2024)



1. Motivation

Table 1. Results of Gleason group prediction. (mean ± standard)

Method	Acc↑	AUC↑	QWK↑	F1↑
CE	28.08 ± 2.33	53.91 ± 0.61	0.03 ± 0.01	20.62 ± 1.49
FocalLoss [7]	30.70 ± 1.81	55.02 ± 1.54	0.06 ± 0.03	22.82 ± 1.89
EMD [6]	**31.47** ± 2.10	57.84 ± 2.58	**0.08** ± 0.04	25.03 ± 2.18
ExpLoss [9]	31.15 ± 0.62	56.78 ± 1.89	**0.08** ± 0.01	24.67 ± 1.45
OrdinalEncoding [8]	29.49 ± 1.60	-	0.06 ± 0.02	22.91 ± 2.38
Softlabel [8]	25.13 ± 3.66	56.19 ± 3.40	0.05 ± 0.03	22.39 ± 3.36
Ours	29.35 ± 1.82	**60.06** ± 1.22	0.07 ± 0.02	**25.58** ± 2.25

1. Motivation

CSAW-M: An Ordinal Classification Dataset for Benchmarking Mammographic Masking of Cancer

Moein Sorkhei[*1, 2] Yue Liu[*1, 2] Hossein Azizpour[1]
Edward Azavedo[3,4] Karin Dembrower[3,5] Dimitra Ntoula[4]
Athanasios Zouzos[3,4] Fredrik Strand[3,4] Kevin Smith[1, 2]

[1]KTH Royal Institute of Technology, Stockholm, Sweden
[2]SciLifeLab, Stockholm, Sweden
[3]Karolinska Institutet, Stockholm, Sweden
[4]Karolinska University Hospital, Stockholm, Sweden
[5]Saint Göran Hospital, Stockholm, Sweden

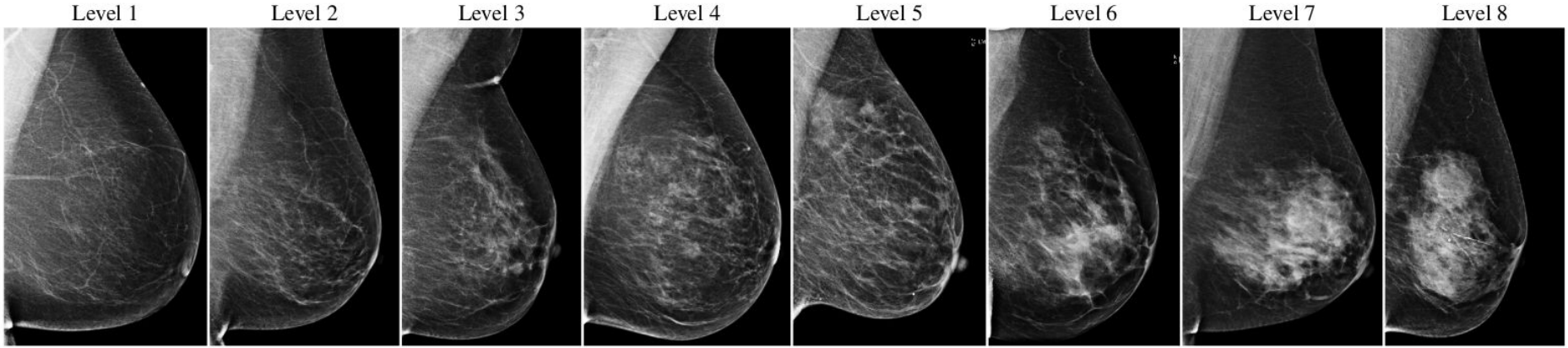


Figure 1: Masking potential, or the possibility for cancer to be obscured in a mammogram, from CSAW-M. From left to right, the most agreed-upon images among five expert annotators for level 1 (lowest masking potential, easist to assess) to level 8 (highest masking potential, hardest to assess).

1. Motivation

Table 3: Comparison of expert and model performance on ordinal classification of masking potential for the public test set. Mean and standard deviation of 5 runs are reported for the models.

		Kendall's τ_b ↑	AMAE ↓	F_1 on level 1-2 ↑	F_1 on level 7-8 ↑
Experts	Expert 1	0.7232	**0.6762**	0.7940	0.6154
	Expert 2	0.7279	0.7167	0.7465	**0.6316**
	Expert 3	0.5450	1.0037	0.7363	0.5200
	Expert 4	0.5554	1.0390	0.5430	0.6242
	Expert 5	0.6342	1.0321	0.6885	0.5225
Models	One-hot	0.7126 ± 0.0083	0.8108 ± 0.0145	0.7855 ± 0.0136	0.5950 ± 0.0243
	Multi-hot	**0.7625 ± 0.0030**	0.7086 ± 0.0142	**0.8064 ± 0.0188**	0.5571 ± 0.0320

AMAE: Average Mean Absolute Error

Kendall's τ: Correlation between rankings

1. Motivation

BI-RADS Classification of Calcification on Mammograms

Yanbo Zhang[1], Yuxing Tang[1], Zhenjie Cao[1], Mei Han[1], Jing Xiao[2], Jie Ma[3], and Peng Chang[1(✉)]

[1] PAII Inc., Palo Alto, USA
[2] Ping An Insurance (Group) Company of China, Shenzhen, China
[3] Department of Radiology, Shenzhen People's Hospital, Shenzhen, China



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Table 1. A comparison of evaluation metrics on two classifiers. Consider an example list of six samples with various BI-RADS categories, all samples are correctly predicted except for the one with BI-RADS 4A. Both calcification example cases 1 and 2 are with the BI-RADS category 4A, and final confirmed as malignant and benign, respectively.

Cases	Classifiers	Predicted BI-RADS	QWK	MAQWK	
				MAQWK-M	MAQWK-B
Case 1 (4A, Malignant)	Classifier 1	2	0.9024	0.8356	–
	Classifier 2	4C	0.8919	**1.0**	–
Case 2 (4A, Benign)	Classifier 1	4C	0.8919	–	0.8153
	Classifier 2	2	0.9024	–	**1.0**

2. How NOT to Assess Ordinal Models

Grades: None, Mild, Severe (0, 1, 2)

True \ Pred	None	Mild	Severe
None	8	2	0
Mild	0	5	0
Severe	0	2	3

Accuracy = 16/20 = 80%

QWK = 0.83

2. How NOT to Assess Ordinal Models

Grades: None, Mild, Severe (0, 1, 2)

True \ Pred	None	Mild	Severe
None	8	1	1
Mild	0	5	0
Severe	0	2	3

Accuracy = 16/20 = 80% QWK = 0.72

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

$$\mathrm{QWK} = 1 - \frac{\sum_{i,j} W_{i,j} O_{i,j}}{\sum_{i,j} W_{i,j} E_{i,j}}$$

Where O is a matrix of observations, E is the matrix of expected observations if predictions were at random but preserving class frequency, and W is a cost matrix for mistakes.

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

Tr \ Pr	N	M	S
N	8	1	1
M	0	5	0
S	0	2	3

Need to build Observed and Expected matrices

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

Tr \ Pr	N	M	S
N	8	1	1
M	0	5	0
S	0	2	3

Need to build Observed and Expected matrices

O is what actually happened. Entry Oij is #cases whose true grade is i and predicted grade is j.

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

Tr \ Pr	N	M	S
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Assessing categorical predictions: the queen

Tr \ Pr	N	M	S
N	8	1	1
M	0	5	0
S	0	2	3

Need to build Observed and Expected matrices

= O is what actually happened. Entry Oij is #cases whose true grade is i and predicted grade is j.

E is what would happen by chance, given same marginals.

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1
0	5	0
0	2	3

E is what would happen by chance, given same marginals.

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1	10
0	5	0	5
0	2	3	5
8	8	4	20

E is what would happen by chance, given same marginals.

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1	10
0	5	0	5
0	2	3	5
8	8	4	20

E is what would happen by chance, given same marginals.

$10 \cdot {}^{8}\!/_{20}$	$10 \cdot {}^{8}\!/_{20}$	$10 \cdot {}^{4}\!/_{20}$	10
$5 \cdot {}^{8}\!/_{20}$	$5 \cdot {}^{8}\!/_{20}$	$5 \cdot {}^{4}\!/_{20}$	5
$5 \cdot {}^{8}\!/_{20}$	$5 \cdot {}^{8}\!/_{20}$	$5 \cdot {}^{4}\!/_{20}$	5

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1	10
0	5	0	5
0	2	3	5
8	8	4	20

E is what would happen by chance, given same marginals.

$10 \cdot 8/20$	$10 \cdot 8/20$	$10 \cdot 4/20$	10
$5 \cdot 8/20$	$5 \cdot 8/20$	$5 \cdot 4/20$	5
$5 \cdot 8/20$	$5 \cdot 8/20$	$5 \cdot 4/20$	5

Note: because the class "None" is more prevalent, we get more expectation in the top row

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1	10
0	5	0	5
0	2	3	5
8	8	4	20

E is what would happen by chance, given same marginals.

4	4	2	10
2	2	1	5
2	2	1	5

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1
0	5	0
0	2	3

E =

4	4	2
2	2	1
2	2	1

W =

0	1	2
1	0	1
2	1	0

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

O =

8	1	1
0	5	0
0	2	3

E =

4	4	2
2	2	1
2	2	1

W =

0	1	4
1	0	1
4	1	0

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

$O =$

8	1	1
0	5	0
0	2	3

$O \odot W =$

0	1	4
0	0	0
0	2	0

$E =$

4	4	2
2	2	1
2	2	1

$W =$

0	1	4
1	0	1
4	1	0

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

$O =$

8	1	1
0	5	0
0	2	3

$O \circ W =$

0	1	4
0	0	0
0	2	0

$E =$

4	4	2
2	2	1
2	2	1

$E \circ W =$

0	4	8
2	0	1
8	2	0

$W =$

0	1	4
1	0	1
4	1	0

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

$O =$

8	1	1
0	5	0
0	2	3

$O \odot W =$

0	1	4
0	0	0
0	2	0

$\Sigma = 7$

$E =$

4	4	2
2	2	1
2	2	1

$E \odot W =$

0	4	8
2	0	1
8	2	0

$\Sigma = 25$

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

$O =$

8	1	1
0	5	0
0	2	3

$O \circ W =$

0	1	4
0	0	0
0	2	0

$\Sigma = 7$

QWK = 1-7/25

$E =$

4	4	2
2	2	1
2	2	1

$E \circ W =$

0	4	8
2	0	1
8	2	0

$\Sigma = 25$

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

$O =$

8	1	1
0	5	0
0	2	3

$O \circ W =$

0	1	4
0	0	0
0	2	0

$\Sigma = 7$

QWK = 0.72

$E =$

4	4	2
2	2	1
2	2	1

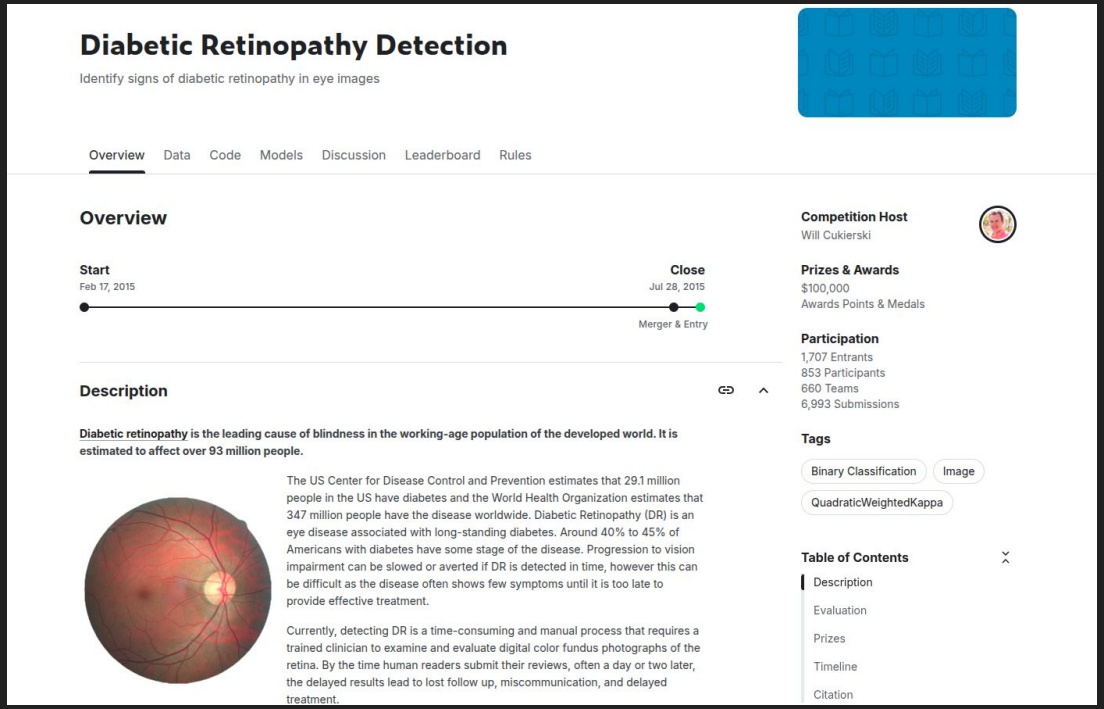
$E \circ W =$

0	4	8
2	0	1
8	2	0

$\Sigma = 25$

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen



Diabetic Retinopathy Detection

Identify signs of diabetic retinopathy in eye images

Overview Data Code Models Discussion Leaderboard Rules

Overview

Start
Feb 17, 2015

Close
Jul 28, 2015

Merger & Entry

Description

Diabetic retinopathy is the leading cause of blindness in the working-age population of the developed world. It is estimated to affect over 93 million people.

The US Center for Disease Control and Prevention estimates that 29.1 million people in the US have diabetes and the World Health Organization estimates that 347 million people have the disease worldwide. Diabetic Retinopathy (DR) is an eye disease associated with long-standing diabetes. Around 40% to 45% of Americans with diabetes have some stage of the disease. Progression to vision impairment can be slowed or averted if DR is detected in time, however this can be difficult as the disease often shows few symptoms until it is too late to provide effective treatment.

Currently, detecting DR is a time-consuming and manual process that requires a trained clinician to examine and evaluate digital color fundus photographs of the retina. By the time human readers submit their reviews, often a day or two later, the delayed results lead to lost follow up, miscommunication, and delayed treatment.

Competition Host
Will Cukierski

Prizes & Awards
$100,000
Awards Points & Medals

Participation
1,707 Entrants
853 Participants
660 Teams
6,993 Submissions

Tags
Binary Classification
Image
QuadraticWeightedKappa

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3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

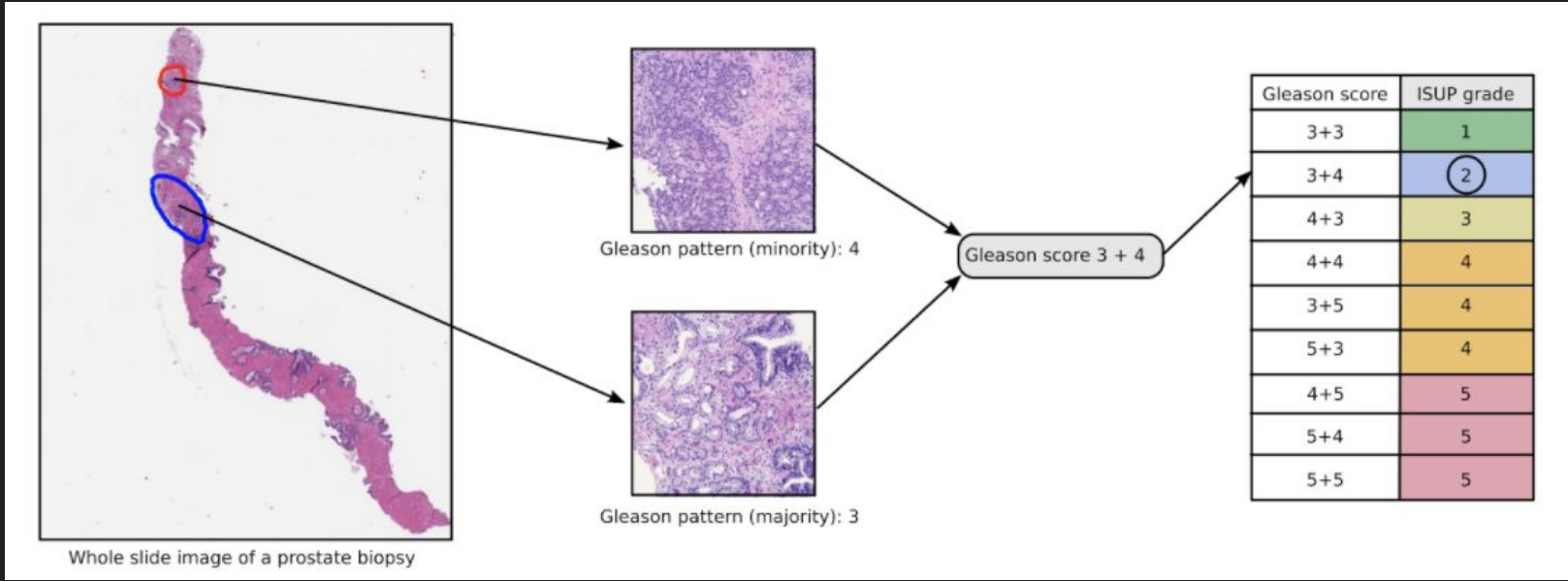
Public Private

This leaderboard is calculated with approximately 20% of the test data. The final results will be based on the other 80%, so the final standings may be different.

#	Team	Members	Score	Entries	Last
1	Min-Pooling		0.86035	37	11y
2	o_O		0.85425	61	11y
3	Reformed Gamblers		0.85106	58	11y
4	Julian de Wit & Daniel Hammack		0.85105	49	11y

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

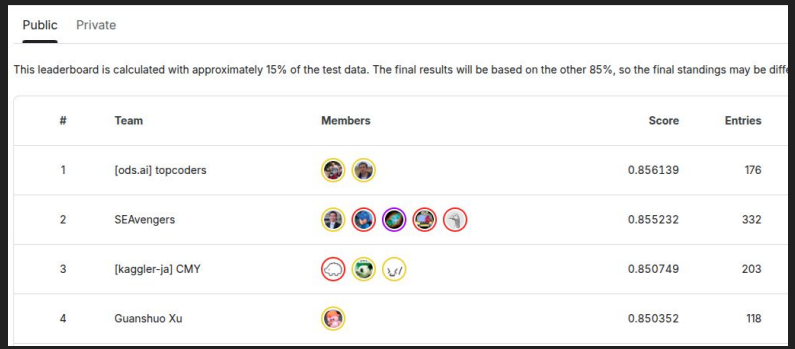


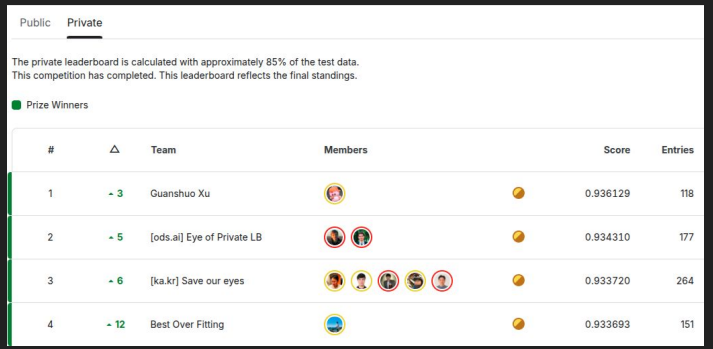
Prostate cANcer graDe Assessment (PANDA) Challenge

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen



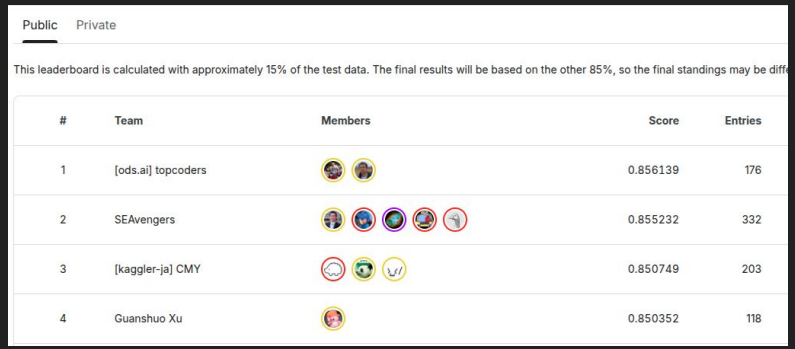


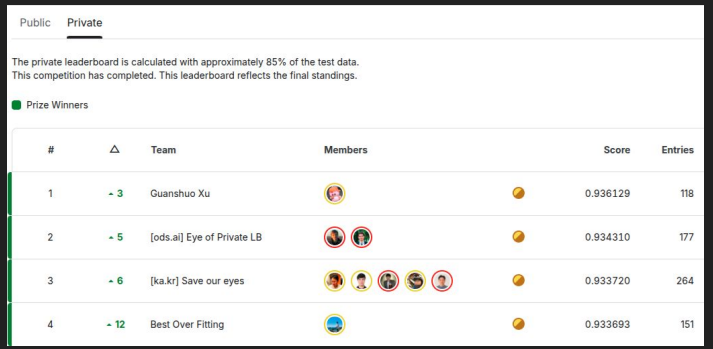


3. How to Assess Ordinal Models

Assessing categorical predictions: the queen







3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

Imagine a doctor who:

- **If the patient is None: 80% says None, 10% Mild, 10% Severe**
- **If Mild: always says Mild**
- **If Severe: 60% says Severe, 40% says Mild**

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

Imagine a doctor who:

- **If the patient is None: 80% says None, 10% Mild, 10% Severe**
- **If Mild: always says Mild**
- **If Severe: 60% says Severe, 40% says Mild**

For a population of 10 None, 5 Mild, 5 Severe out of 20, we are at QWK=0.72, Accuracy=80%, MAE=0.25

3. How to Assess Ordinal Models

Assessing categorical predictions: the queen

Imagine a doctor who:

- **If the patient is None: 80% says None, 10% Mild, 10% Severe**
- **If Mild: always says Mild**
- **If Severe: 60% says Severe, 40% says Mild**

For a population of 10 None, 5 Mild, 5 Severe out of 20,
we are at QWK=0.72, Accuracy=80%, MAE=0.25

For a population of 100 None, 5 Mild, 5 Severe out of 110,
we are at QWK=0.37, Accuracy=80%, MAE=0.29

3. How to Assess Ordinal Models

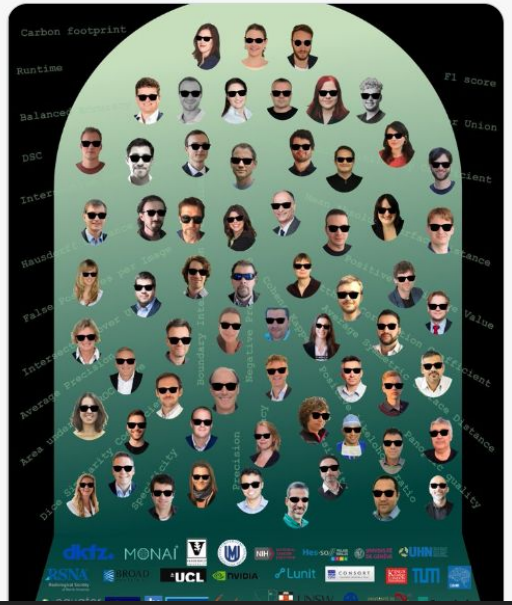
Lena Maier-Hein @lena_maierhein · Jun 6, 2022

UNCOVERING problems is good, SOLVING them is even better.

To address pitfalls of #imageanalysis validation (arxiv.org/abs/2104.05642) we have compiled the

🕶️METRICS RELOADED framework🕶️: arxiv.org/abs/2206.01653

Now challenge us, help us, join us! onlinesurvey.dkfz.de/index.php/4543... 1/



Lena Maier-Hein @lena_maierhein · Jun 6, 2022

Get involved: Join us on the journey by giving feedback via this questionnaire (very short responses possible!): onlinesurvey.dkfz.de/index.php/4543.... Contributions will be acknowledged; substantial ones with a co-authorship. @naturemethods @MiccaiStudents @MICCAI_Society @ELLISforEurope 4/

1 2 15

3. How to Assess Ordinal Models

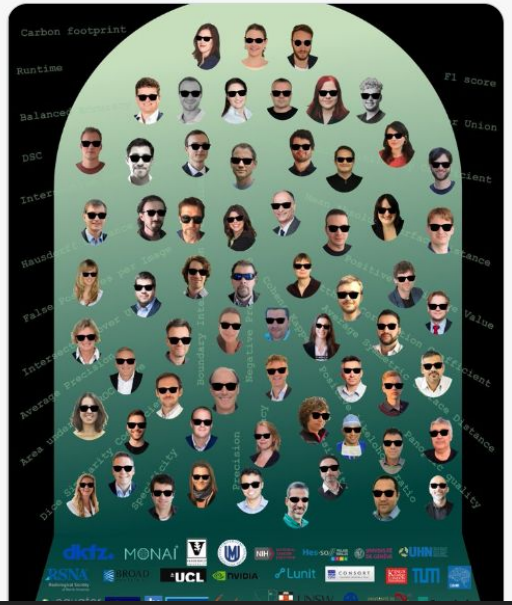
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1 2 15

Metrics Reloaded: Invitation Inbox ×

Reinke, Annika <a.reinke@dkfz-heidelberg.de> 28 Jul 2022, 09:00 Reply

to me, Lena, Paul

Dear Adrian,

we would like to personally thank you for your extensive feedback in our Metrics Reloaded community survey! Your feedback was very valuable and we are currently discussing how to best incorporate your valid points. Especially, we would like to thank you for bringing up the pitfall for the quadratic Cohen's Kappa. We are currently discussing whether or not to add this metric to the framework, given the limitations (e.g. the prevalence dependency). Anyways, we would like to incorporate the pitfall into our pitfalls paper and would like to add you to the co-authors for this paper (https://arxiv.org/abs/2104.05642), if you agree.

Furthermore, we would like you to offer to become a member of our Metrics Reloaded expert consortium (https://arxiv.org/abs/2206.01653). If you agree, this would mean to be part of the final Delphi process in reaching a consensus on the suggested recommendations, to help us finalizing the biomedical use cases, for which we instantiate the framework and to proof-read the final draft.

3. How to Assess Ordinal Models

Assessing categorical predictions: alternatives

- Regression metrics: MSE, MAE

[MSE is actually the numerator in QWK, $\sum W \cdot O = N \cdot MSE$]

- Correlation metrics: e.g. Kendall Tau

- Expected Cost

3. How to Assess Ordinal Models

Assessing categorical predictions: alternatives

EXPECTED COST (EC)/NORMALIZED EC (ECN)

Synonyms: Expected prediction error, Expected loss

$$EC = w_{miss} \cdot \underbrace{\frac{FN}{TP+FN}}_{P_{miss}} \cdot \underbrace{\frac{TP+FN}{TP+TN+FP+FN}}_{P_{tar}} + w_{FA} \cdot \underbrace{\frac{FP}{TN+FP}}_{P_{FA}} \cdot \underbrace{1 - \frac{TP+FN}{TP+TN+FP+FN}}_{P_{tar}}$$

$$= w_{miss} \cdot \frac{\square}{\square} \cdot \frac{\square}{\square} + w_{FA} \cdot \frac{\square}{\square} \cdot 1 - \frac{\square}{\square}$$

P_{miss}: FN (miss) rate, P_{FA}: FP (false alarm) rate
P_{tar}: prior probability (prevalence)
w_{miss}/w_{FA}: (estimation of) costs of the respective errors; can be adjusted as a weighting of them.

VALUE RANGE: $[0, \infty) \downarrow$
EC can be assumed to be positive if costs are non-negative, which can be done without loss of generality.

DESCRIPTION

EC is a generalization of the probability of error (which is, in turn, 1 - Accuracy) for cases in which errors cannot all be considered to have equally severe consequences. It is defined as the expectation of the cost, where the cost incurred on a certain sample depends on the sample's class and the decision made for that sample. In practice, the expectation can be estimated as a simple average of the costs over the evaluation samples.
EC describes the weighted sum of error rates. It can be used to measure discrimination and calibration in one score.

VARIANT

Normalized EC (ECN): normalizes EC by the EC of a naive system.

3. How to Assess Ordinal Models

Assessing categorical predictions: my pov

We should report qwk/EC
and then bal. accuracy/auc at
clinically meaningful endpoints

3. How to Assess Ordinal Models

Assessing probabilistic predictions

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3. How to Assess Ordinal Models

Assessing probabilistic predictions

- Proper Scoring Rules

Measure discrimination+calibration at individual item level

Most popular: Brier Score, Negative Log-Likelihood

$$\text{Brier}(\mathbf{p}, \mathbf{y}) = ||\mathbf{p} - \mathbf{y}||_2^2 \qquad \text{NLL}(\mathbf{p}, \mathbf{y}) = -\log(\mathbf{p}_y)$$

Example: $y = 3,\ \mathbf{y} = (0, 0, 1), \mathbf{p}_{\text{bad}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \mathbf{p}_{\text{better}} = \left(0, \frac{1}{3}, \frac{2}{3}\right)$

$\text{Brier}(\mathbf{p}_{\text{bad}}, \mathbf{y}) = 2/3$ $\quad \text{Brier}(\mathbf{p}_{\text{better}}, \mathbf{y}) = 2/9$ $\quad \text{Brier}(\mathbf{y}, \mathbf{y}) = 0$

$\text{NLL}(\mathbf{p}_{\text{bad}}, \mathbf{y}) \approx 0.477$ $\quad \text{NLL}(\mathbf{p}_{\text{better}}, \mathbf{y}) \approx 0.176$ $\quad \text{NLL}(\mathbf{y}, \mathbf{y}) = 0$

3. How to Assess Ordinal Models

Assessing probabilistic predictions

Property 1 A Scoring Rule is **proper** (PSR) if its value is minimal when the probabilistic prediction coincides with the ground-truth in expectation. **Example**: The Brier Score, $\text{Brier}(\mathrm{p}, \mathrm{y}) = ||\mathrm{p} - \mathrm{y}||_2^2$ is always non-negative, and it decreases to 0 when p = y.

Property 2 A PSR is **local** if its value is only dependent on the probability assigned to the correct category. **Example**: The Log. Score $\mathcal{L}(\mathrm{p}, \mathrm{y}) = -\log(\mathrm{p}_c)$ aka Cross-Entropy, is Local, as it only looks at the correct class c. The Brier Score is non-local.

Property 3 A PSR is **sensitive to distance** if probability placed in categories further away from the correct class is more penalized.

3. How to Assess Ordinal Models

Assessing probabilistic predictions

The Brier and the Logarithmic Scores are insensitive to distance

Definition: The RPS is the squared ℓ_2 distance between the cumulative distributions Y of the target y, and P of the probabilistic prediction p, discounting the last component (=1) and normalizing to unit interval:

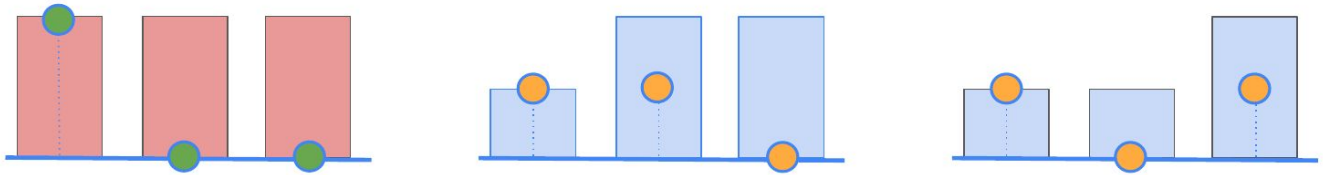
$$\mathrm{RPS}(p, y) = \frac{1}{K} \sum_{i=1}^{K-1} |P - Y|_2^2 = \frac{1}{K} \sum_{i=1}^{K-1} \sum_{j=1}^{i} (p_j - y_j)^2$$

3. How to Assess Ordinal Models

Assessing probabilistic predictions

$$\mathrm{RPS}(\mathbf{p}, \mathbf{y}) = \frac{1}{K} \sum_{i=1}^{K-1} |\mathbf{P} - \mathbf{Y}|_2^2 = \frac{1}{K} \sum_{i=1}^{K-1} \sum_{j=1}^{i} (p_j - y_j)^2$$

$\mathbf{y} = [1, 0, 0]$ $\quad$ $\mathbf{p_1} = [1/2, 1/2, 0]$ $\quad$ $\mathbf{p_2} = [1/2, 0, 1/2]$



$\mathbf{Y} = [1, 1, 1]$ $\quad$ $\mathbf{P_1} = [1/2, 1, 1]$ $\quad$ $\mathbf{P_2} = [1/2, 1/2, 1]$

$$\mathrm{RPS}(\mathbf{p_1}, \mathbf{y}) = \frac{1}{2}\|\mathbf{P_1} - \mathbf{Y}\|_2^2 = \frac{1}{8} < \frac{1}{4} = \mathrm{RPS}(\mathbf{p_2}, \mathbf{y}) = \frac{1}{2}\|\mathbf{P_2} - \mathbf{Y}\|_2^2$$

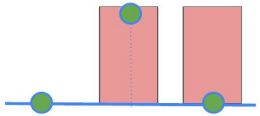
3. How to Assess Ordinal Models

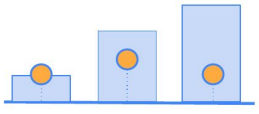
Assessing probabilistic predictions

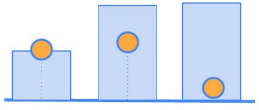
Extension: we want quadratic growth with distance to correct cat.

$$\text{sa-RPS}(\mathbf{p}, \mathbf{y}) = \frac{1}{K}\left[\sum_{i=1}^{K-1}\left|\sum_{j=1}^{i}(p_j - y_j)^2\right|\right]^2$$

$\mathbf{y_2} = [0, 1, 0]$ $\quad$ $\mathbf{p_1} = [3/10, 4/10, 3/10]$ $\quad$ $\mathbf{p_2} = [4.5/10, 5/10, 0.05/10]$







$\mathbf{Y_2} = [0, 1, 1]$ $\quad$ $\mathbf{P_1} = [3/10, 7/10, 1]$ $\quad$ $\mathbf{P_2} = [4.5/10, 9.5/10, 1]$

$$\text{RPS}(\mathbf{p_1}, \mathbf{y}) = \frac{1}{2}\left(\left[\frac{3}{10}\right]^2 + \left[\frac{-3}{10}\right]^2\right) = 0.09 \quad < \quad 0.1025 = \text{RPS}(\mathbf{p_1}, \mathbf{y}) = \frac{1}{2}\left(\left[\frac{-4.5}{10}\right]^2 + \left[\frac{0.05}{10}\right]^2\right)$$

$$\text{sa-RPS}(\mathbf{p_1}, \mathbf{y}) = \left[\frac{1}{2}\left(\left|\frac{3}{10}\right| + \left|\frac{-3}{10}\right|\right)\right]^2 = 0.09 \quad > \quad 0.0625 = \text{sa-RPS}(\mathbf{p_1}, \mathbf{y}) = \left[\frac{1}{2}\left(\left|\frac{-4.5}{10}\right| + \left|\frac{0.5}{10}\right|\right)\right]^2$$

3. How to Assess Ordinal Models

Assessing probabilistic predictions

Evaluating Evaluation Metrics - AURSC*:

1) **Sort** test set probabilistic predictions according to a score *S*.

3. How to Assess Ordinal Models

Assessing probabilistic predictions

Evaluating Evaluation Metrics - AURSC*:

1) **Sort** test set probabilistic predictions according to a score *S*.
2) Progressively **remove** samples of worst quality according to *S*.

3. How to Assess Ordinal Models

Assessing probabilistic predictions

Evaluating Evaluation Metrics - AURSC*:

1) **Sort** test set probabilistic predictions according to a score *S*.
2) Progressively **remove** samples of worst quality according to *S*.
3) Take the arg-max on the remaining probabilistic predictions and **compute** Quadratically-Weighted Kappa.

3. How to Assess Ordinal Models

Assessing probabilistic predictions

Evaluating Evaluation Metrics - AURSC*:

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3. How to Assess Ordinal Models

Assessing probabilistic predictions

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5) **Find area under the curve** of removed samples vs performance.

3. How to Assess Ordinal Models

Assessing probabilistic predictions

Evaluating Evaluation Metrics - AURSC*:

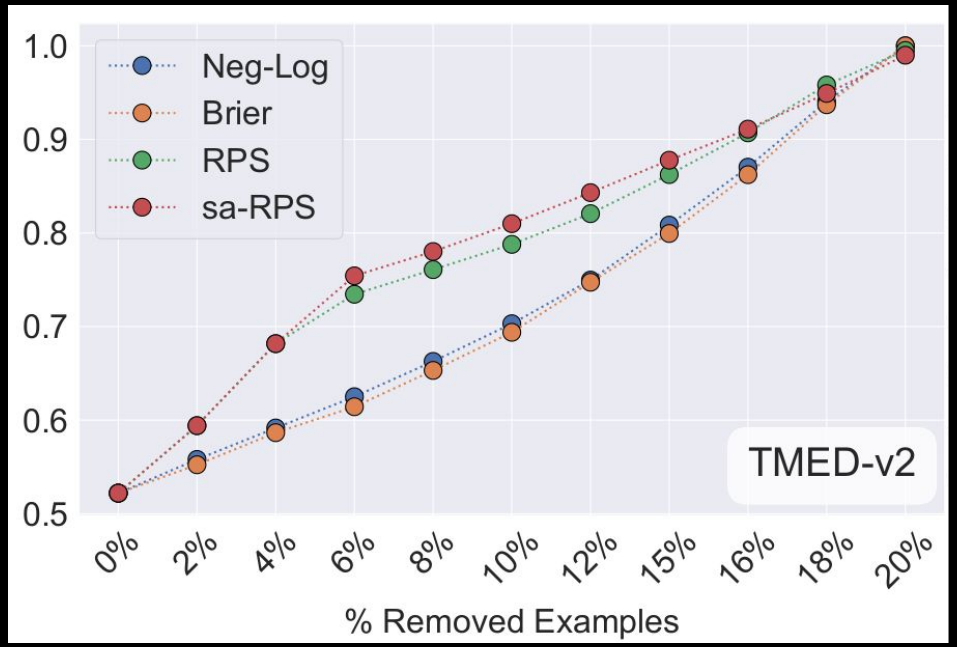
1) **Sort** test set probabilistic predictions according to a score *S*.
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5) **Find area under the curve** of removed samples vs performance.

What we expect to see:

As we remove test set samples considered as worse classified by (SA)-RPS, we expect to more quickly improve QWK. Therefore, the **AURSC** should be higher than when sorting with NLL or the Brier score.

3. How to Assess Ordinal Models

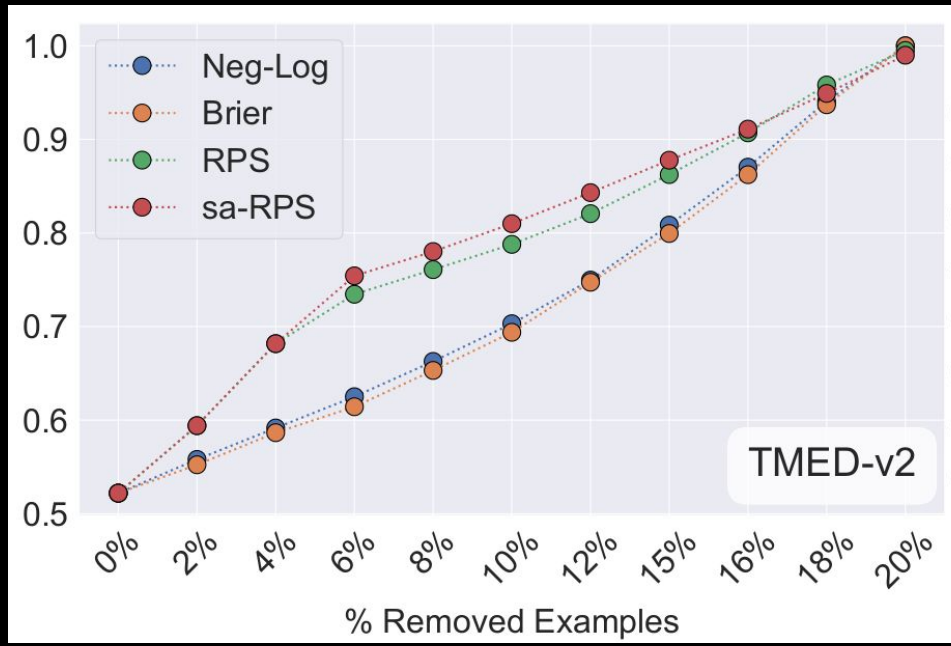
a) T-MED: US video frames of patients with an Aortic Stenosis (AS) diagnostic label from three categories: None, Early AS, or Significant AS.

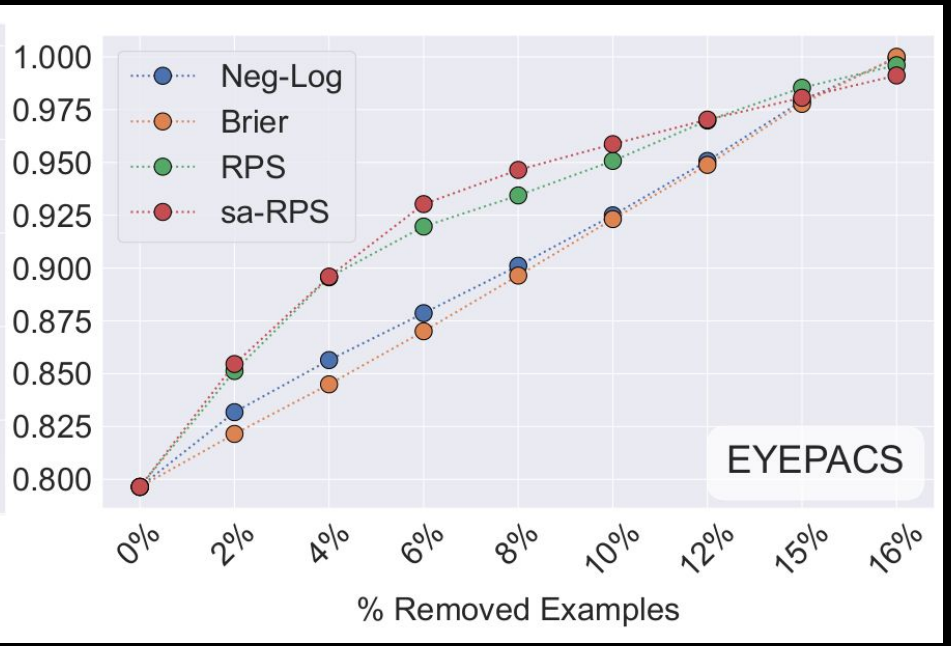


3. How to Assess Ordinal Models

a) **T-MED**: US video frames of patients with an Aortic Stenosis (AS) diagnostic label from three categories: **None**, **Early** AS, or **Significant** AS.

b) **EYEPACS**: Retinal images for DR grading into **5 progressive stages**.





3. How to Assess Ordinal Models

Assessing probabilistic predictions



Medical Image Computing and Computer Assisted Intervention – MICCAI 2023

Performance Metrics for Probabilistic Ordinal Classifiers

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